

Activity 3.1.2 Ohm's Law Lab

PLTW

OHM'S LAW LAB

- ≡ Introduction to Electricity
- ≡ Electrical Circuit Components
- ≡ Electrical Circuit Values
- ≡ Tinkercad Basics
- ≡ Effect of Voltage on Current
- ≡ Measuring Current
- ≡ Resistor Values
- ≡ Ohm's Law
- ≡ Resistance in a Series Circuit

≡ Mystery Resistor

≡ Conclusion

Introduction to Electricity

GOALS	MATERIALS	RESOURCES
• Build functioning electrical circuits. • Model functioning electrical circuits using Autodesk® Tinkercad®. • Test circuit parameters using a digital multimeter. • Determine the mathematical relationship between voltage, current, and resistance (Ohm's Law).		
GOALS	MATERIALS	RESOURCES
• Variable power supply • Breadboard • 100 Ω resistors • 330 Ω resistors		

- 1,000 Ω resistors
- Mystery resistors
- LEDs
- Hookup wires
- Digital multimeter

GOALS	MATERIALS	RESOURCES
• <u>Ohm's Law Data Collection (Microsoft Excel)</u> • <u>Ohm's Law Data Collection (Google Sheets)</u> • <u>Tinkercad® Circuit Tips</u> • <u>Multimeter Operation</u> • <u>Breadboards</u> • <u>Activity 3.1.2 Ohm's Law Lab (Downloadable PDF)</u>		

Electricity is a form of energy that results from the flow of charged particles including protons and electrons. [Electrical energy](#) can be transformed into many useful forms, such as light, heat, sound, and

motion. Electrical energy can be converted, stored, transmitted, and reconverted efficiently into other forms of energy. Figures 1 and 2 show examples of electricity that you may be familiar with.



Figure 1. Static Electricity



Figure 2. Solar Panels

Electrical energy has three properties: current, voltage, and resistance.

Term	Definition
current	The net transfer of electric charge (electron movement along a path) per unit of time. The symbol is I and the unit is Ampere which abbreviates to A.
voltage	The potential difference measured in volts. The amount of work to be done to move a charge from one point to another along an electric

	circuit. The symbol is V and unit is Volt which abbreviates to V.
resistance	The opposition that a device or material offers to the flow of direct current. The symbol is R and the unit is Ohm which abbreviates to Ω .

Understanding the relationship among current, voltage, and resistance enables engineers to design efficient, safe, and functional electrical circuits. Electrical circuits are made up of:

- An energy source to provide voltage
- Conductors to allow current to travel
- Insulators to limit current travel
- A load, or the component that is consuming electrical power

Electrical circuits provide an uninterrupted path for current travel in two distinct categories of design: [series circuits](#) and [parallel circuits](#).

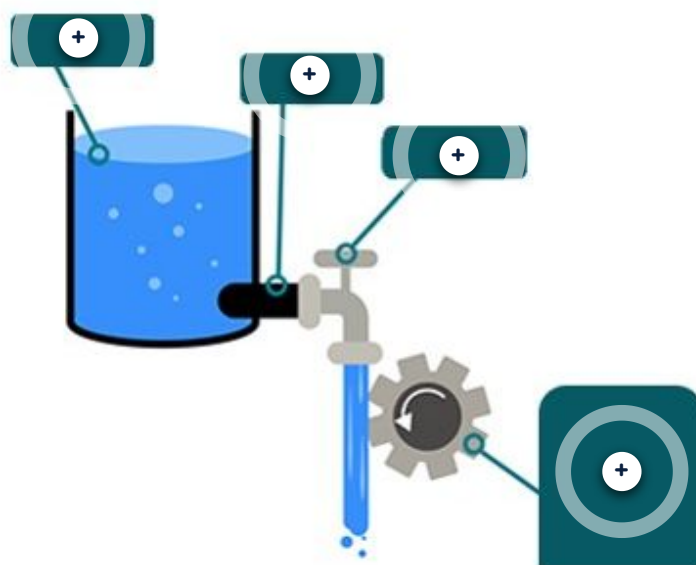
How Does Electricity Work?

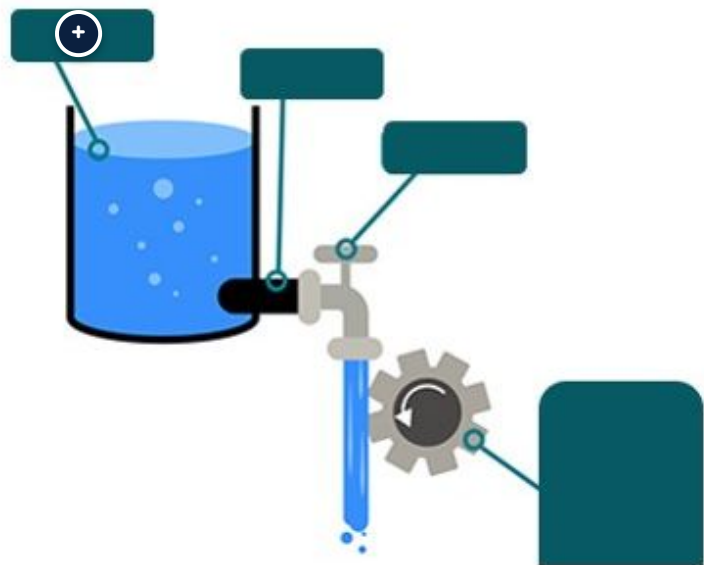
The Electricity table shows the equivalent elements of water systems and electrical circuits.

Electricity	
Water System	Electrical Circuit
Water is stored in a tank.	Electrons are stored in a battery.

Water flows through pipes.	Electrons flow through wires.
Water flow is stopped or allowed by a faucet valve.	Electron flow is stopped or allowed by a switch.
Potential energy of water can be converted into kinetic energy through a water wheel.	Electrical energy can be converted into light and thermal energy through a lightbulb.

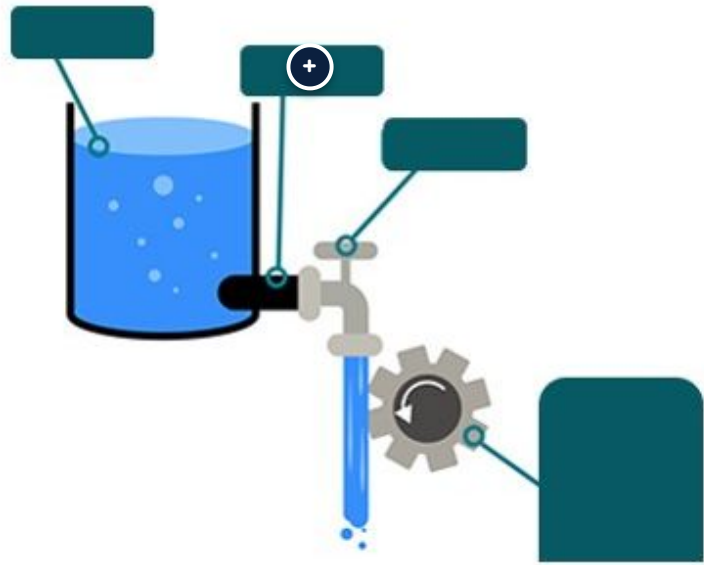
A household water supply system can provide an analogy to understand an electrical system. Review the Water System Analogy interactive graphic below to learn more.





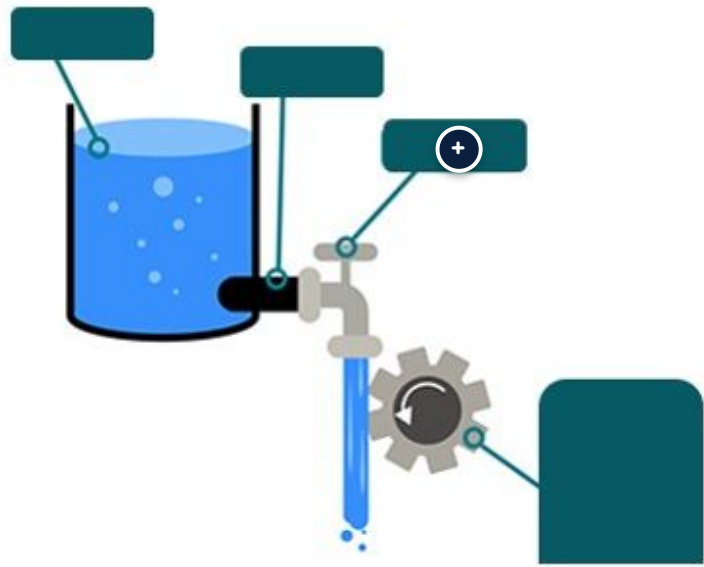
Tank

A tank of water acts as a battery.



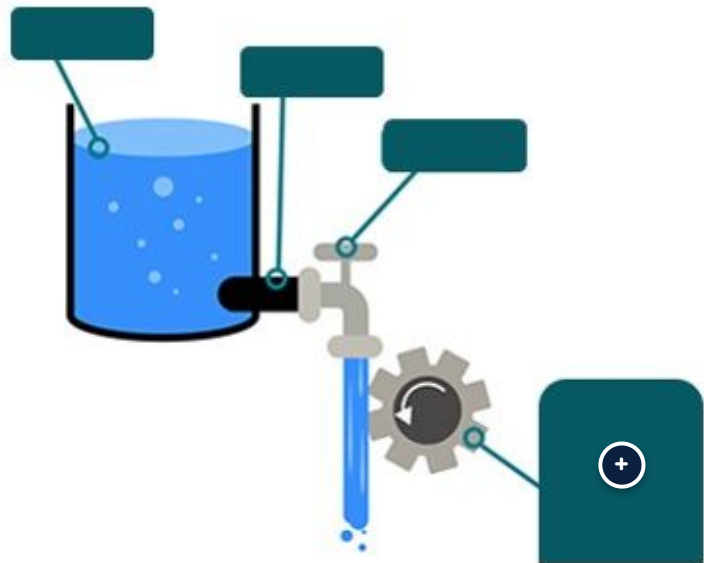
Pipe

A pipe attached to the water tank acts as a wire.



Faucet

A faucet attached to the pipe acts as a switch for turning the water on or off.



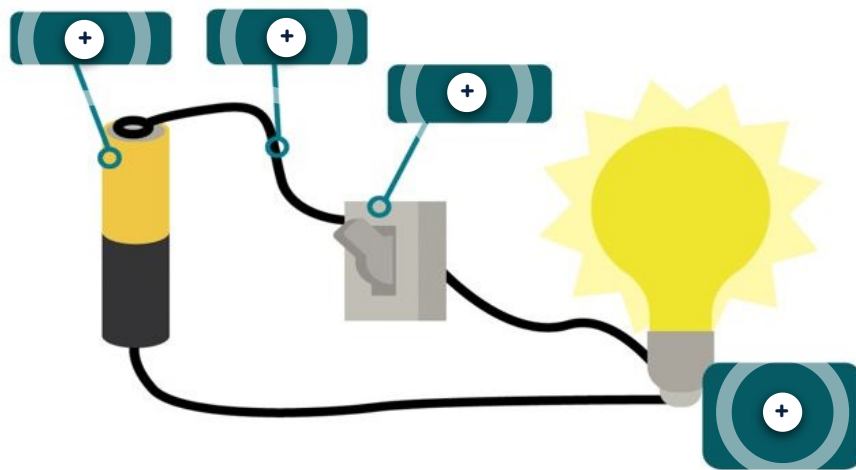
Types of Energy

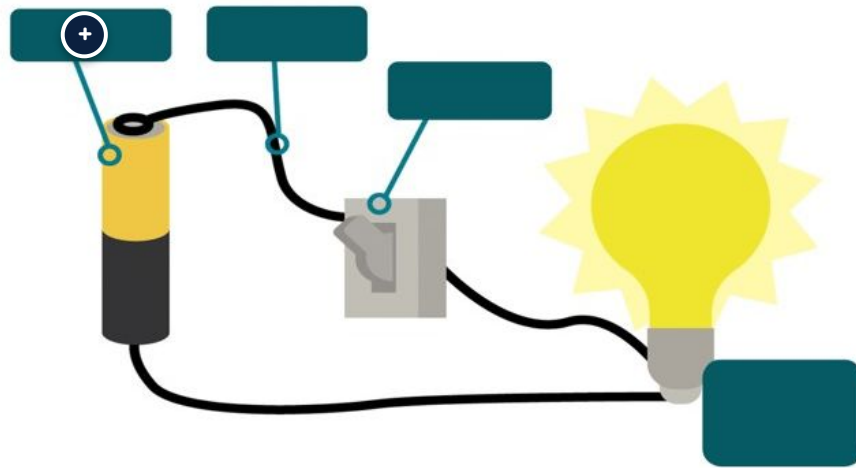
When the flow of water is turned on, linear mechanical energy is transformed into rotational mechanical energy.

Water System Analogy interactive graphic

Electrical Circuit Components

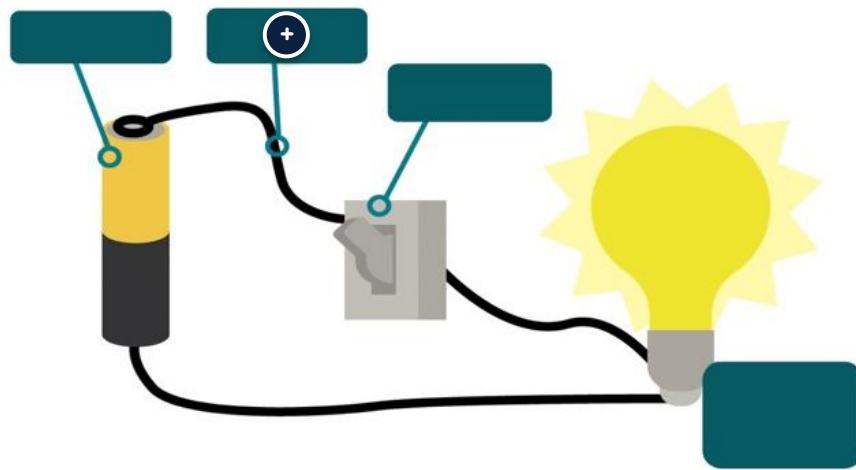
All [electrical circuits](#) have these basic components: source, conductor, and load, and sometimes, a switch. The electrons must have a continuous path from the source through the circuit back to the source. Review the Electrical Circuit Components interactive graphic to learn more.





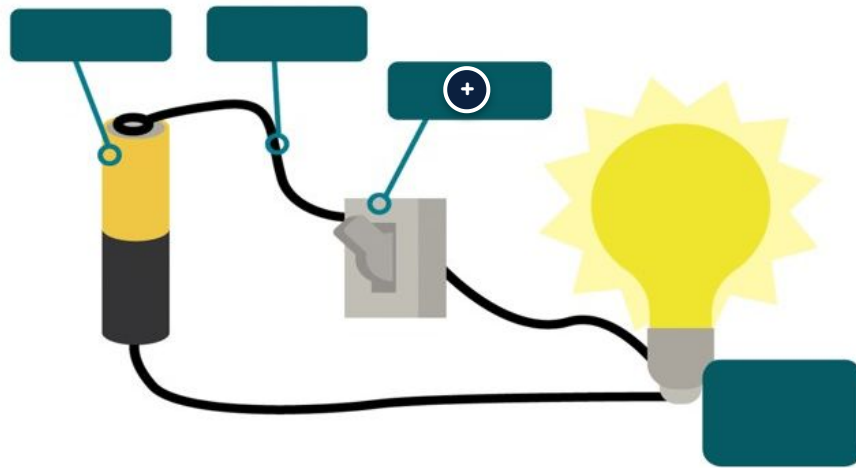
Battery

A battery acts as a tank.



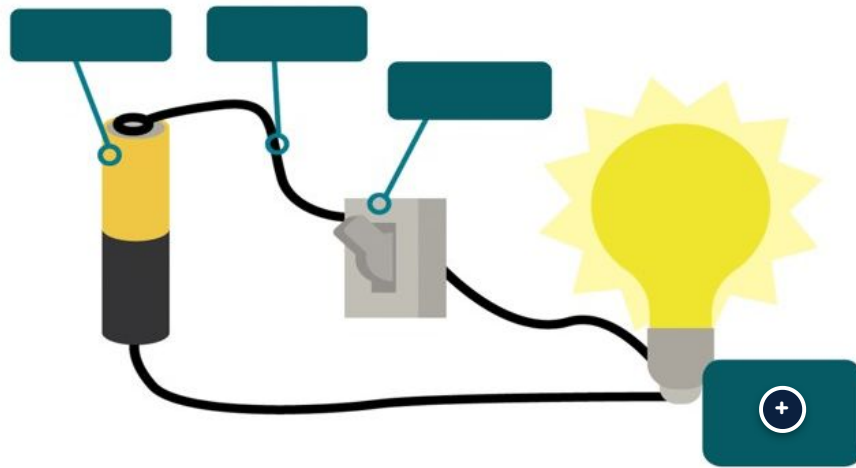
Wire

A wire connected to the positive end of the battery acts as a pipe.



Switch

A light switch connected to the wire acts as a faucet. When the light switch is turned on, the light is too.



Types of Energy

When the light switch is on, electrical energy is transformed into light and thermal energy.

Electrical Circuit Components interactive graphic

Electrical circuit diagrams use industry-specific symbols. Figure 3 shows a circuit similar to the circuit in the Electrical Circuit Components interactive graphic as a schematic with the appropriate circuit symbols.

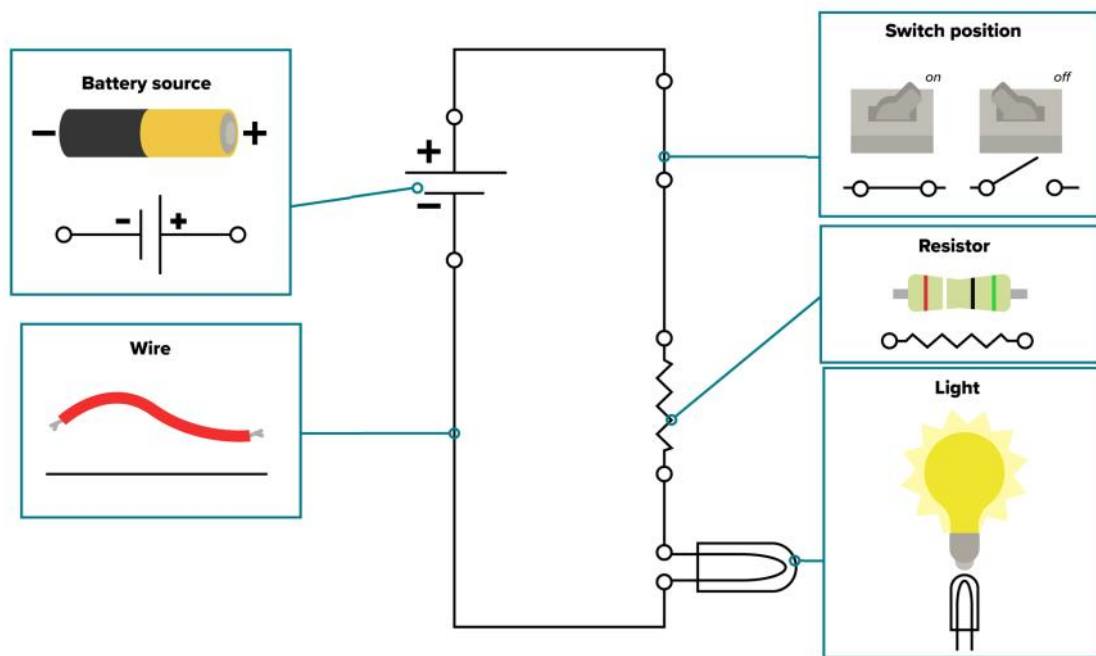


Figure 3. Electrical Circuit Schematic

Electrical Circuit Values

An electrical circuit has several parameters:

- Voltage is thought of as electrical pressure. Greater voltage means greater pressure, meaning the ability to push more current through the circuit.

EXAMPLE: A small toy car's battery has between 1.5 and 3 V to propel it forward, whereas a modern electric vehicle's battery has between 400 and 800 V. The full-sized vehicle needs that greater electrical pressure between battery terminals to provide adequate current to the electric motors.

- Current is the rate of flow of charged particles through a conductor. Larger diameter wires can carry more current, or more electrons per second.
- Electrical resistance is a measure of opposition to current flow. Current decreases as resistance increases. Engineers use resistors to limit the amount of current through sensitive electrical components so as not to overload them.

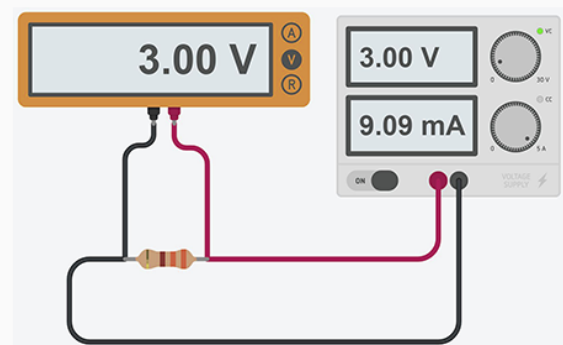
Property	Abbreviation	Unit	Symbol
Voltage	V	Volt	V
Current	I	Ampere	A
Resistance	R	Ohm	Ω

Measuring Circuit Values

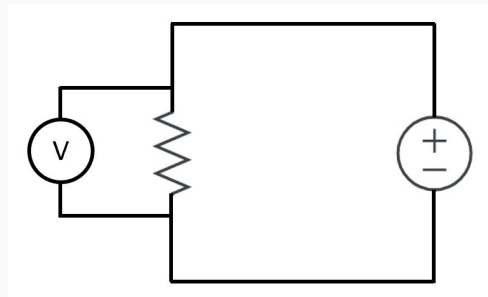
A digital [multimeter](#) is a tool that can measure a variety of circuit parameters including voltage, current, and resistance. When you use a multimeter to measure current, the multimeter must be in series with the circuit.

You can set up this tool to measure voltage across circuit elements, current through circuit branches, and resistance of individual circuit components like resistors. The flashcards show how to connect to a multimeter to measure circuit values and how they are represented in a circuit diagram.

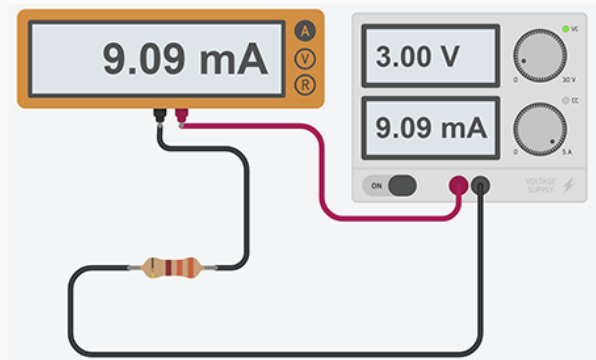
Voltmeter Simulated
Configuration



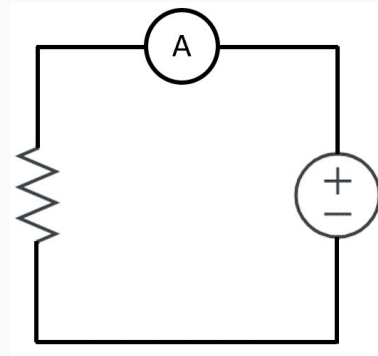
Voltmeter Schematic Symbols



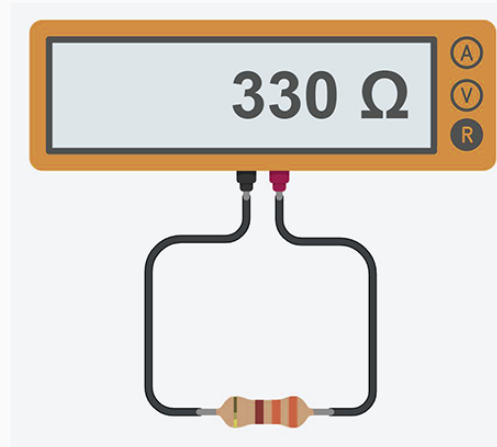
Ammeter Simulated Configuration



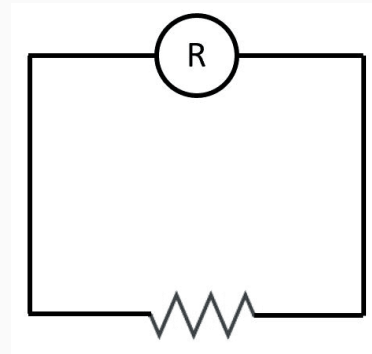
Ammeter Schematic Symbols



Ohmmeter Simulated Configuration



Ohmmeter Schematic Symbols



See [Multimeter Operation](#) for more detailed instructions on how to use a multimeter to measure circuit values.

Tinkercad Basics

Throughout this activity, you will construct electrical circuits using a breadboard and you will model circuits using Autodesk® Tinkercad®. Reference the Tinkercad Basics video along with [Tinkercad](#) [Circuit Tips](#) and [Breadboards](#) topics as needed.

Tinkercad Basics video

Effect of Voltage on Current

1

Using Tinkercad®, build a circuit that connects a lightbulb to a power supply, as shown in Figure 4.

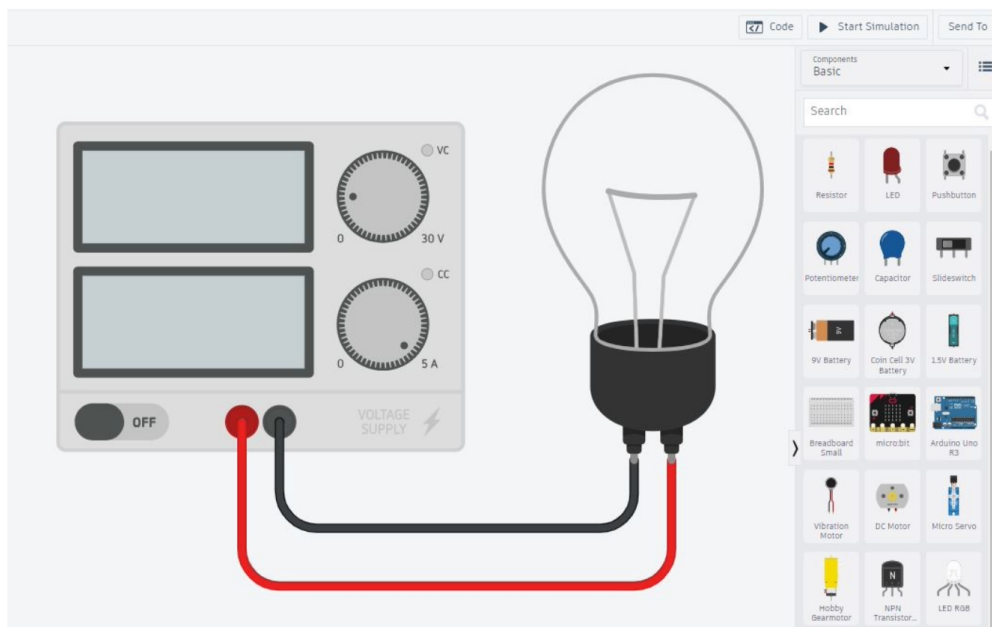


Figure 4. Tinkercad Circuit Off

2

In the top right of the Tinkercad window, select **Start Simulation**. The circuit will turn on, as shown in Figure 5.

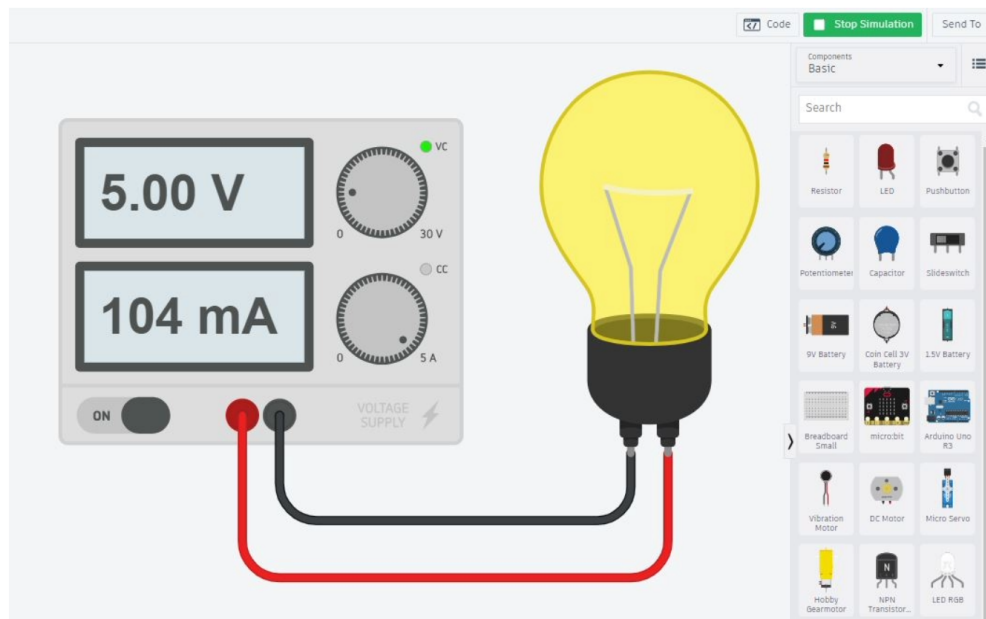


Figure 5. Tinkercad Circuit On

3

Click and rotate the voltage dial on the power supply to change the voltage to a value between 1 and 4 V. Note the reading for current (in milliamps) on the power supply. Record the current at that voltage.

4

Change the voltage to a value between 8 and 12 V. Record the current at that voltage.

Turn and Talk



How does changing the voltage affect the current?

Check Your Response

As voltage increases, current increases.

5

Adjust the voltage up and down between 1 and 12 V. As you change the voltage, observe the brightness of the bulb.

Turn and Talk



How does changing the voltage affect the bulb brightness?

Check Your Response

As voltage increases, bulb brightness increases.

Measuring Current

To continue with this activity, you must have a working knowledge of breadboards and multimeters. Review [Breadboards](#) and [Multimeter Operation](#) as necessary.

6

Observe the circuit in Figure 6.

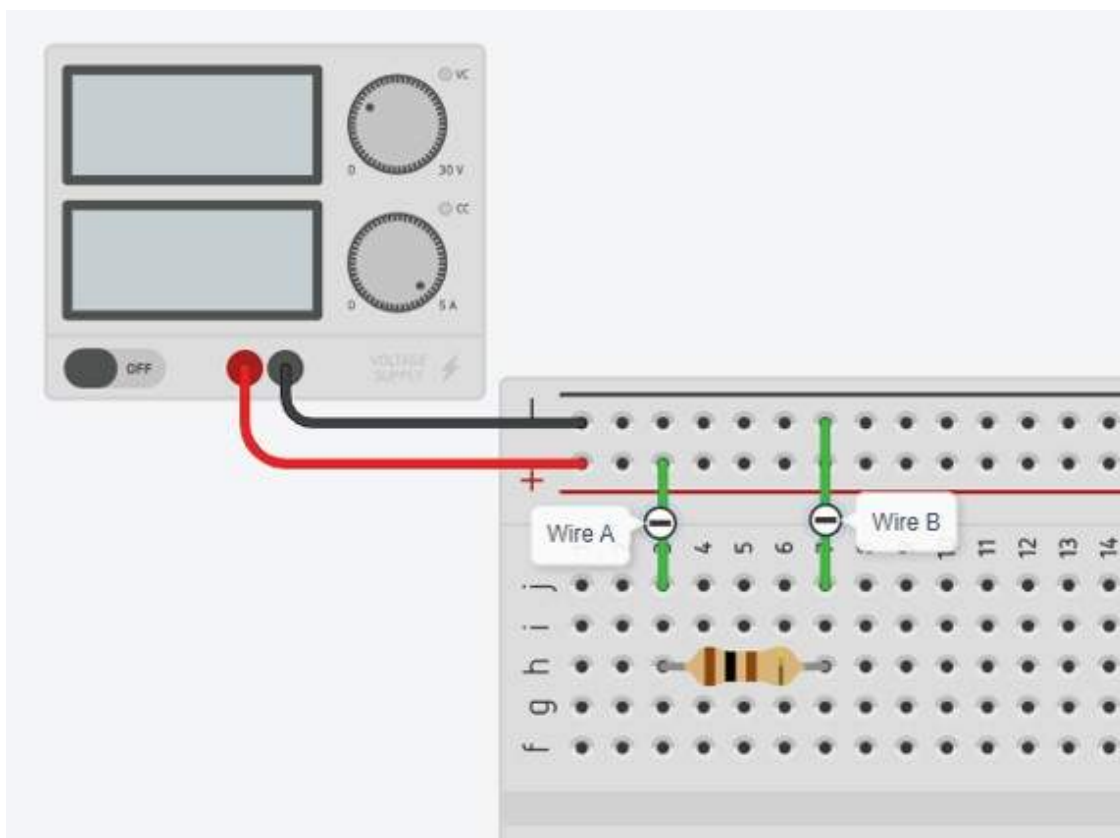


Figure 6. Circuit with Single Resistor

Turn and Talk



Describe the path of the current through the circuit, starting with the positive terminal of the power supply. (Conventional current flows from positive to negative.)

Check Your Response

- From the positive terminal of the power supply through the red wire to the positive rail of the breadboard.
- From the positive rail of the breadboard to hole j3 on the breadboard through the green wire labeled Wire A.
- From hole j3 to h3 through the internal rail of the breadboard (not visible).
- From hole h3 to hole h7 through the resistor.
- From hole h7 to j7 through the internal rail of the breadboard (not visible).
- From hole j7 to the negative rail of the breadboard through the green wire labeled Wire B.
- From the negative rail of the breadboard through the black wire to the negative terminal of the power supply.

7

In Tinkercad, build the circuit from Figure 6.

- Set the voltage to 3 V and use a 100 Ω resistor.

- Wire A should connect the resistor to the positive (+) rail.
- Wire B should connect the resistor to the negative (-) rail.

8

Position a digital multimeter under your power supply, as shown in Figure 7.

- Connect the positive (+) rail of the breadboard to the red terminal of the multimeter.
- Connect the black terminal of the multimeter to the resistor.

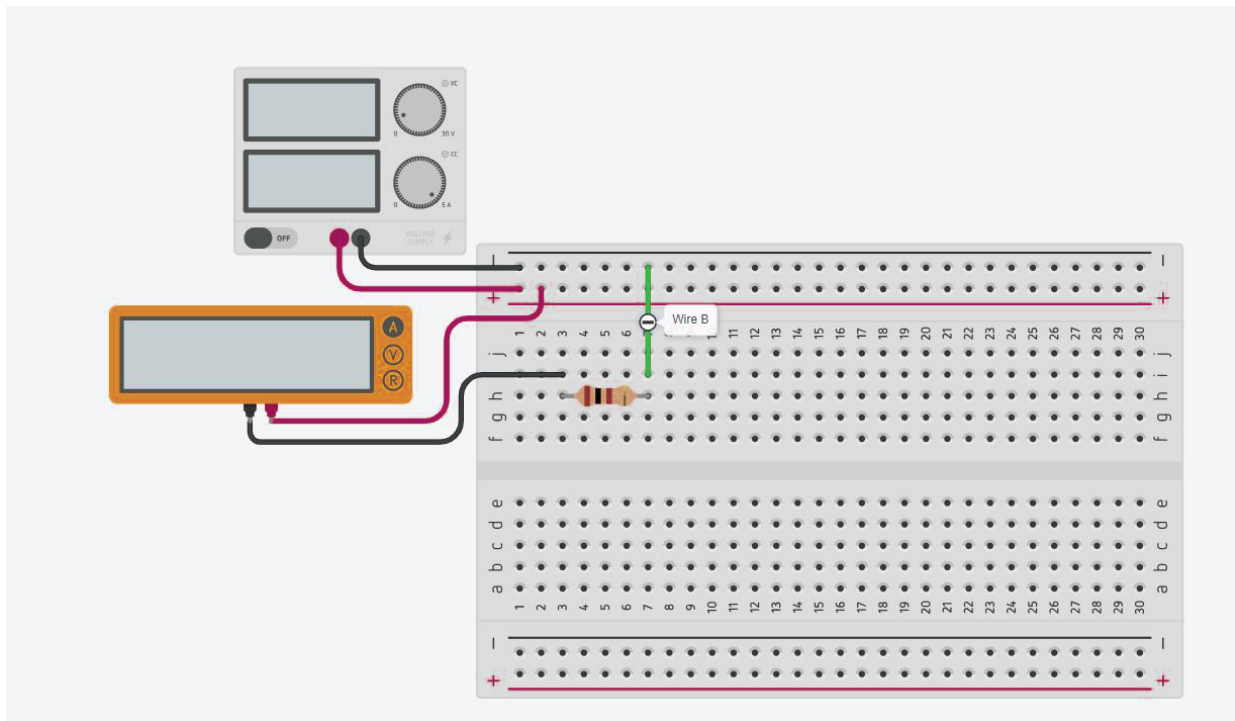


Figure 7. Measuring current at Wire A

9

Position a digital multimeter above your breadboard, to the right of the power supply, as shown in Figure 8.

- Connect the red terminal of the multimeter to one end of the resistor.
- Connect the black terminal of the multimeter to the negative rail of the breadboard.

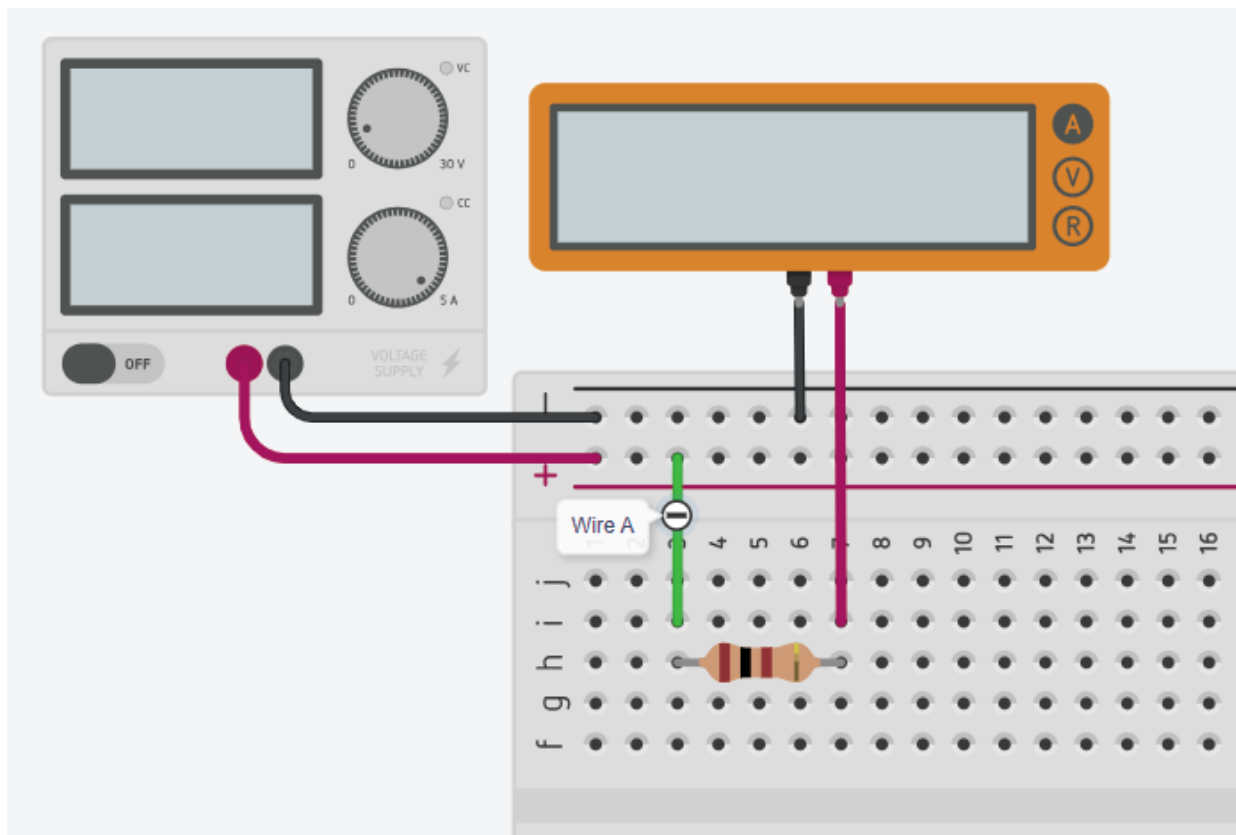


Figure 8. Measuring current at Wire B

Turn and Talk



Was your prediction correct? What can you say about the current through a simple circuit with only one path for current to flow?

Check Your Response

The current is the same throughout this circuit.

Resistor Values

You may have noticed that as you changed a resistor's value in Tinkercad®, the color striping on the resistor changed. These colors indicate the [nominal](#) resistance and [tolerance](#) values of the resistor. To determine the resistance of individual resistors, you can use a digital multimeter set as an ohmmeter, or a resistor color guide like that shown in Figure 9.

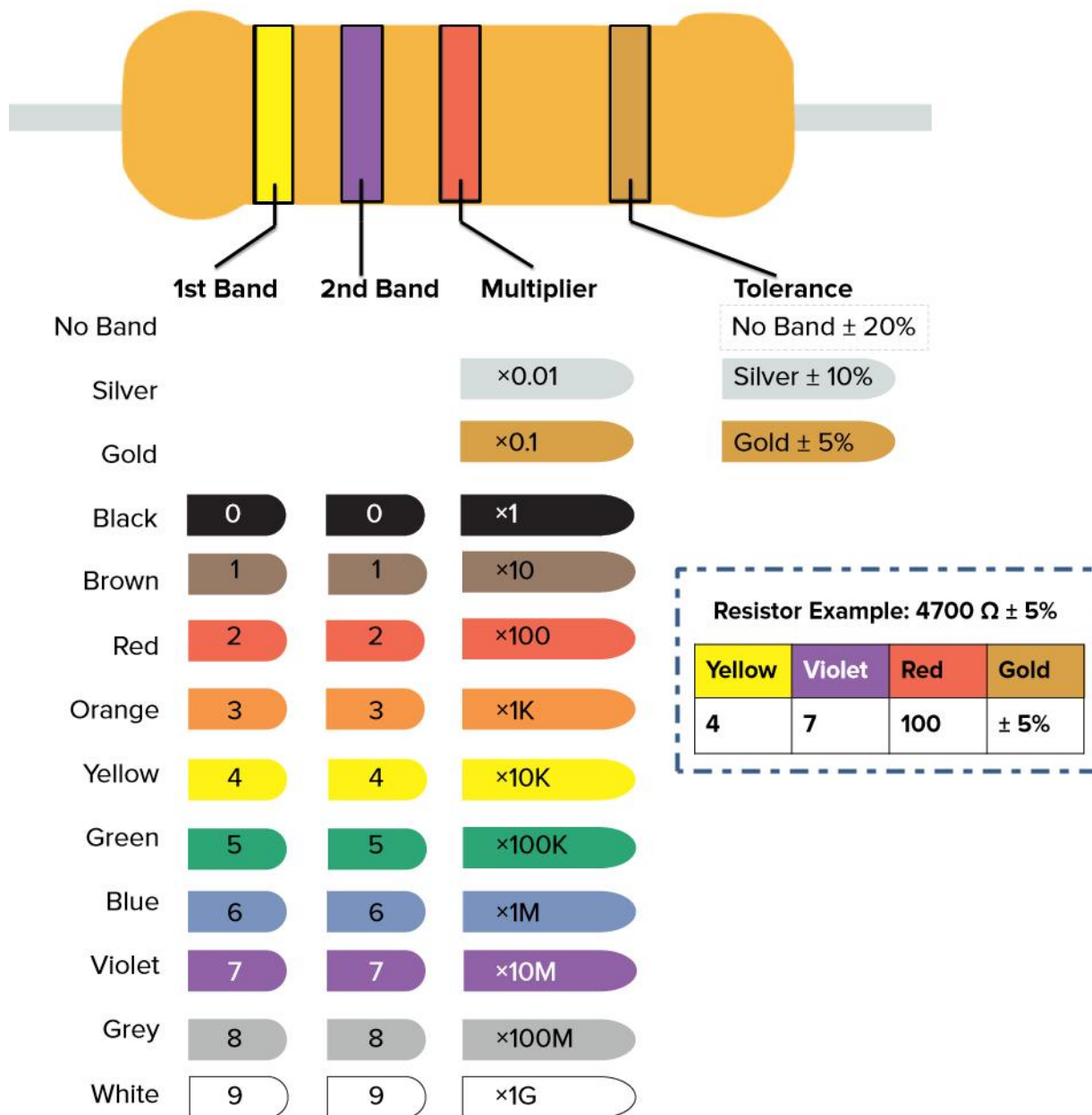


Figure 9. Resistor Color Guide

- The first and second bands (yellow and violet in this example) indicate the first and second digits (4 and 7) of the resistor's value. Those digits are the first two of the resistor's value.

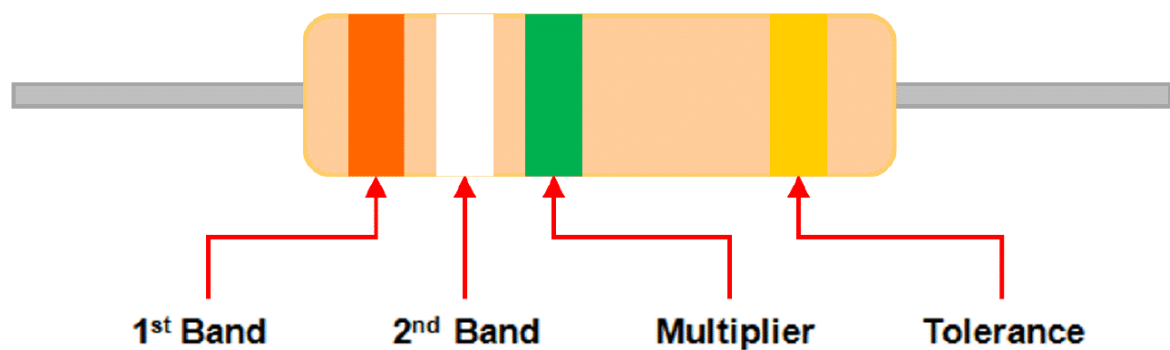
- The third band indicates a multiplier. Multiply this value by the first two digits. In this example, the multiplier is red, which corresponds to 100 Ω . Multiply that by 47 to get 4700 Ω .
- The fourth band indicates the resistor's tolerance value. In this example, the gold band corresponds to a $\pm 5\%$ tolerance value. So this is a 4,700 Ω resistor with a $\pm 5\%$ tolerance value.

Check Your Understanding

Use Figure 9. Resistor Color Guide to determine the resistance values or colors of the resistors in the following images.

Determine the nominal value for a resistor with:

- 1st Band - Orange
- 2nd Band - White
- Multiplier - Green
- Tolerance - Gold

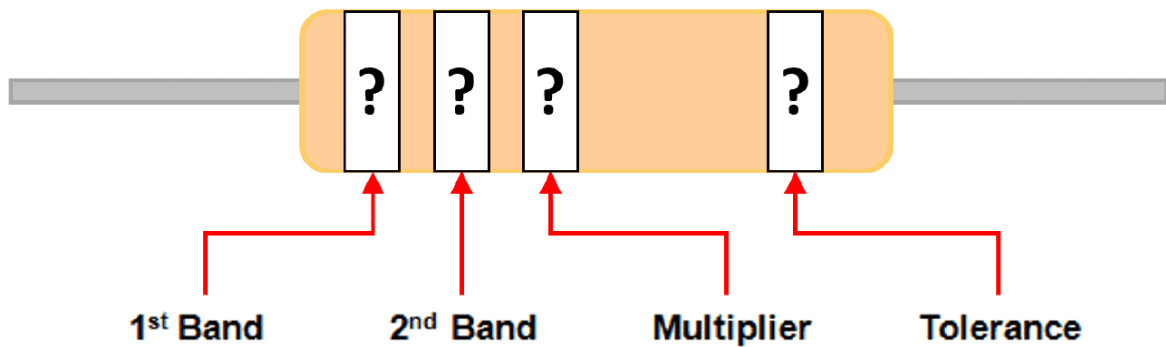


Type your answer here

SUBMIT

NEXT QUESTION

Determine the color bands for a $270\text{ k}\Omega \pm 10\%$ resistor.

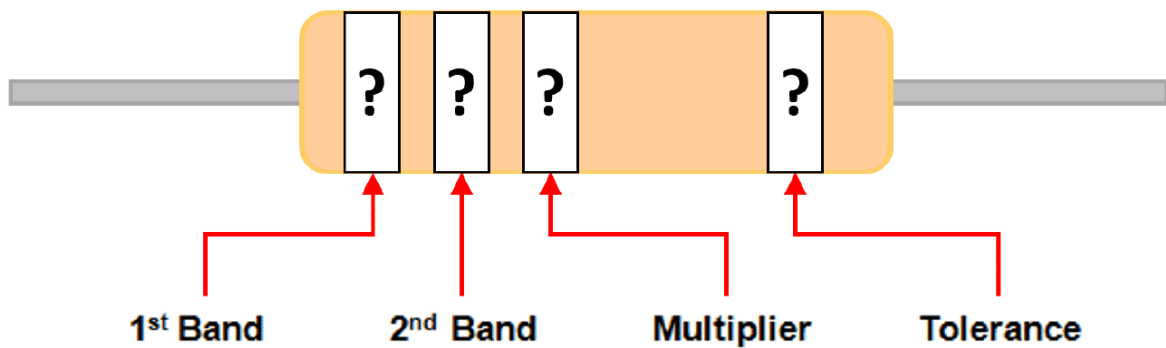


- ☐ Red, Violet, Yellow, Silver
- ☐ Yellow, Blue, Yellow, Silver
- ☐ Red, Violet, Silver, Silver
- ☐ Red, Violet, Yellow, No Band

SUBMIT

NEXT QUESTION

Determine the color bands for a $1.5 \text{ k}\Omega \pm 5\%$ resistor.



- ☐ Gold, Violet, Green, Brown
- ☐ Yellow, Blue, Yellow, Silver
- ☐ Brown, Green, Red, Gold



Brown, Green, Violet, Gold

SUBMIT

COMPLETE CHECK FOR UNDERSTANDING

Ohm's Law

By the end of this investigation, you will arrive at a quantitative relationship between voltage, current, and resistance. To develop an equation, you'll need to collect data.

10

Download and save [Ohm's Law Data Collection \(Microsoft Excel\)](#) or [Ohm's Law Data Collection \(Google Sheets\)](#).

11

Build a physical circuit on the breadboard (not a simulation) with a 1,000 Ω (1 k Ω) resistor.

- Vary the voltage of the power supply from 3 to 9 V in 1.5 V increments
- At each interval, measure the current as shown in Figures 10 and 11.
- In the spreadsheet, record these values in the column titled Current (A).

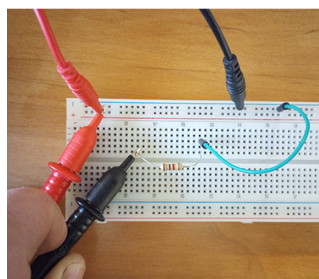
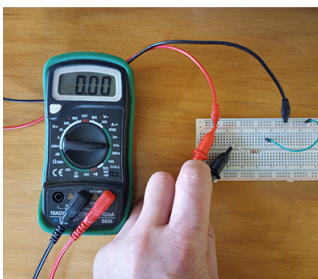


Figure 11. A close-up view

Figure 10. A multimeter set to
20 m

Reflection



Based on the data you collected, what is the mathematical relationship between voltage, resistance, and current? Express this answer as an equation. Review your equation with a partner.

12

Using Tinkercad®, test your equation with multiple combinations of voltage and resistance. Does your equation work in all cases? Modify if necessary.

Turn and Talk



Why is Tinkercad a useful tool for analyzing circuits? Have you identified any shortcomings?

Time to Practice

13

Using your derived equations, solve the following practice problems.

NOTE: You may need to algebraically manipulate your equation depending on what quantity you are solving for.



Engineering Notebook

- A typical mobile phone draws 0.5 A current and has an overall resistance of $8\ \Omega$. How many Volts are required to operate the phone?
- A motorized toy car runs on two AA batteries, providing a total of 3 V. The resistance of the motor in the toy car is $60\ \Omega$. How much current does the toy car draw from the batteries?
- A table lamp is connected to a regular 120 V wall outlet and draws 0.5 A current. Find the bulb's resistance.

Resistance in a Series Circuit

In a series circuit, there is only one path for current to flow. How is overall circuit resistance calculated in a series circuit? You will explore this by building and analyzing circuits with one LED, two LEDs, and three LEDs.

14

Build a circuit on a breadboard with one LED, a $330\ \Omega$ resistor, and the power supply set to 4.5 V. Ensure the circuit has only one path for current to flow.

- Turn on the power supply.
- Observe the LED's brightness. Consider taking a photo of the circuit; you will compare the brightness of this LED to another circuit in step 18.

Turn and Talk



How could you add an LED to your circuit and still have a series circuit? What do you think will happen to the individual LED brightness if you add a second LED? Explain your thinking.

15

Draw a schematic for a circuit with two LEDs, a $330\ \Omega$ resistor, and a power supply set to 4.5 V. Check your schematic with your instructor. When approved, build that circuit.

16

Determine the individual LED brightness when using two LEDs, as compared to just one LED. Consider taking photos of each circuit configuration to compare brightness levels.

Turn and Talk



How do you think the total resistance of a series circuit is calculated? For example, if you had three $100\ \Omega$ resistors, what would the total resistance of the circuit be? Explain your thinking.

17

Build a series circuit with three $100\ \Omega$ resistors and a power supply set to 6 V. Measure the current.

18

Based on the voltage and current, calculate the total resistance of the circuit.

Need Help?

Use Ohm's Law.

19

To confirm this calculation, measure the resistance:

- Remove the two leads from the power supply.
- Replace them with the leads from the multimeter, which is set to measure resistance.
- Allow the value to register on the multimeter.
- Compare the resistance to your calculation.

20

Develop a mathematical rule (equation) to describe the total resistance of a series circuit. Use R_T to indicate total resistance.

Reflection



Why would it be problematic if your home circuits were wired in series? Explain your thinking.

Mystery Resistor

21

Build a series circuit on a breadboard with a $330\ \Omega$ resistor, a mystery resistor, and your power supply set to 3 V.

22

Using what you've learned about the total resistance in a series circuit, perform some measurements and calculations to determine the resistance of the mystery resistor.

NOTE: Use the multimeter to measure the accuracy of the power supply's voltage. Three volts is the minimum range of the power supply, and it can be off by several tenths of a volt. Use the measured value in calculations. **Do NOT** use the multimeter to measure the resistance of the mystery resistor. Use your knowledge of Ohm's Law.

Turn and Talk



Check in with other teams to learn the value of their mystery resistor.

23

Use the multimeter to confirm the resistance of the mystery resistor.

Conclusion

Question 1

Use Ohm's Law to determine the current draw of an electric saw that is connected to a 120 V circuit and has a resistance of $30\ \Omega$.

Question 2

Use Ohm's Law to determine the value of the mystery resistor of the circuit in Figure 12.

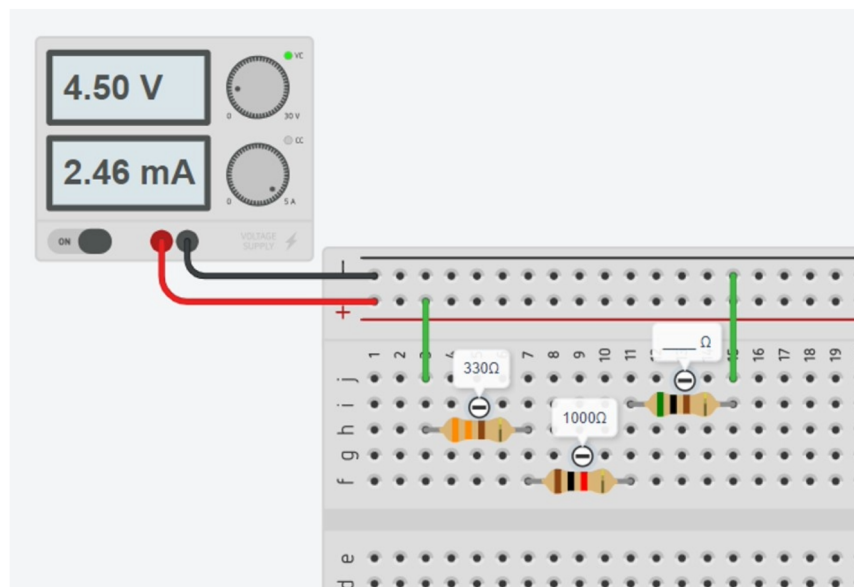


Figure 12. Series Circuit with Three Resistors
